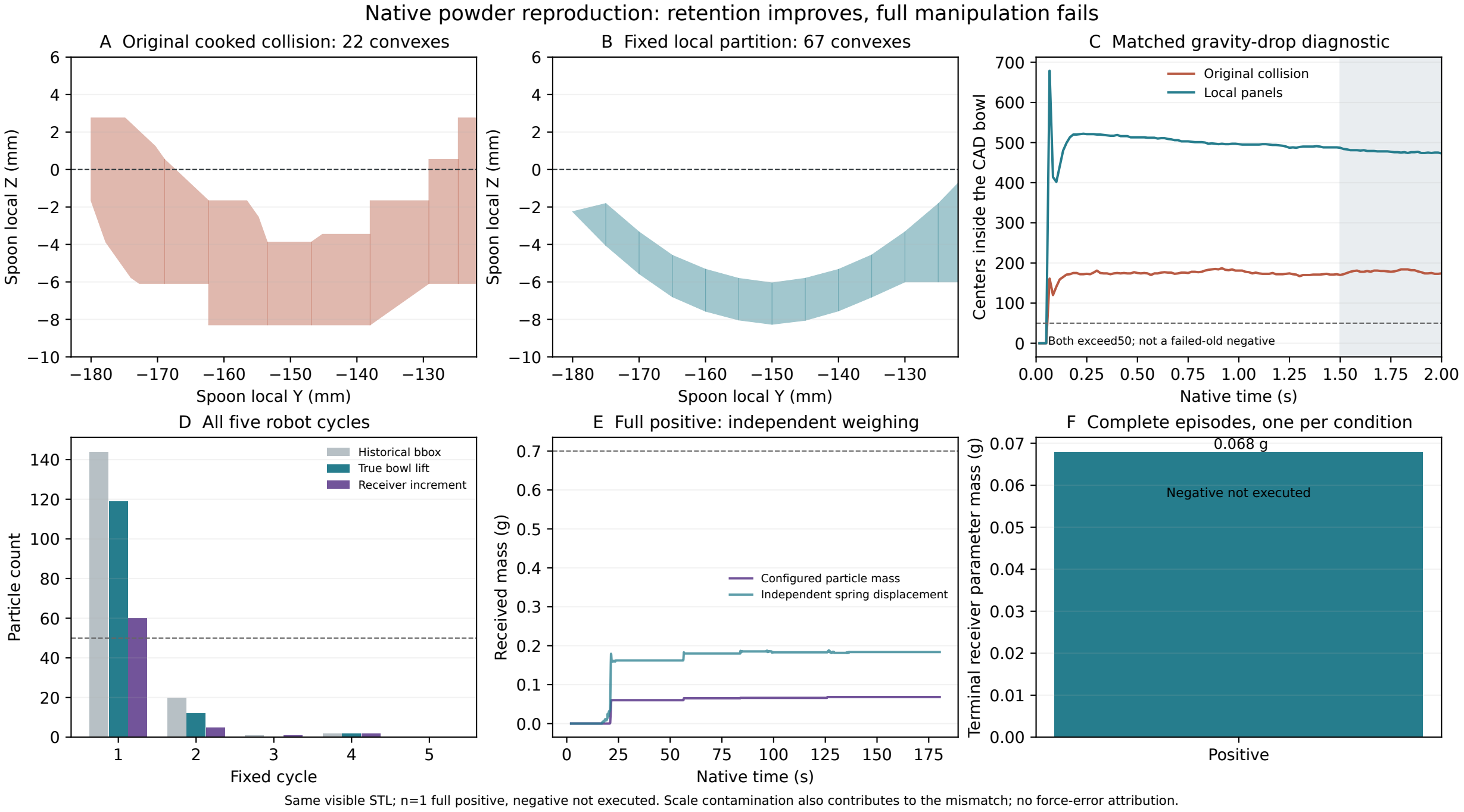
Native powder reproduction: retention improves, full manipulation fails
A Original cooked collision: 22 convexes
Spoon local Z (mm)
Spoon local Y (mm)
B Fixed local partition: 67 convexes
Spoon local Z (mm)
Spoon local Y (mm)
C Matched gravity-drop diagnostic
Original collision
Local panels
Centers inside the CAD bowl
Native time (s)
Both exceed50; not a failed-old negative
D All five robot cycles
Historical bbox
True bowl lift
Receiver increment
Particle count
Fixed cycle
E Full positive: independent weighing
Configured particle mass
Independent spring displacement
Received mass (g)
Native time (s)
F Complete episodes, one per condition
0.068 g
Negative not executed
Terminal receiver parameter mass (g)
Positive
Same visible STL; n=1 full positive, negative not executed. Scale contamination also contributes to the mismatch; no force-error attribution.